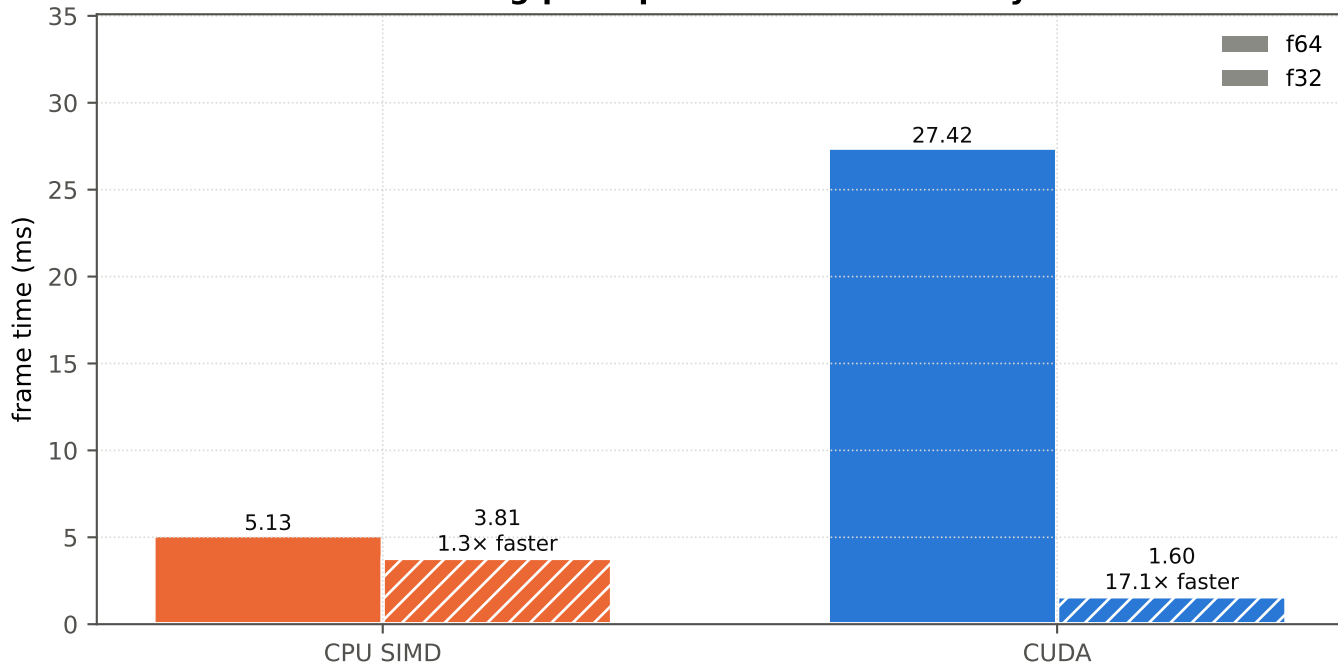


Floating-point precision: what f32 buys



f32 is only NUMERICALLY VALID below zoom 1e6 — above it the mantissa can no longer separate adjacent pixels and the result is fast but wrong. CUDA's f64 penalty is severe because Ampere GeForce runs fp64 at 1/64 the fp32 rate; on the CPU the gap is far smaller. This asymmetry is the whole reason the scheduler has a regime where it wins.